

CAMPBREAK · REQUIREMENTS

CampBreak requirements

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/campbreak		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Amish's round-2 decision 1A (CBK-DDR-004): delivery hose stowed full; R9 now met on paper at 100 m; R6 pull per person 101 N. Requirement wording unchanged
0.4	2026-10-03	Amish Chadha	R9 result after Amish's decision 1A (CBK-DDR-003): check foot valve, suction hose left coupled, go-when-two-arrive rule; delivery hose fill time added. R6 and R11 results updated. Requirement wording unchanged
0.3	2026-10-03	Amish Chadha	TRL 3 with the decisions of CBK-DDR-001 and CBK-DDR-002 applied and results from CBK-CAL-001
0.2	2026-10-03	Amish Chadha	Populated to TRL 2 with concept media, components and first-order numbers
0.1	2026-09-30	Amish Chadha	Initial scaffold

CampBreak requirements

Measurable requirements for one block kit, with the TRL 3 result from the calculation note CBK-CAL-001. Ten are met on paper or by design and R2 can only be shown in a cooking trial at TRL 4. **R9 is now met on paper** for a shelter 100 m from the station, with 3 s to spare.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification (TRL 4 or later)	TRL 3 result	Status
R1	Alarm detects a fast temperature rise	Alarms at a rise of 8 °C (14 °F) per minute or faster, plus a 57 °C fixed threshold	Heat chamber test with logged temperature ramp	Threshold set in firmware, averaged over 60 s	Met by design
R2	Alarm ignores normal cooking	No alarm during 1 hour of stove cooking at 1 m distance	Cooking trial in a mock shelter	Not calculable on paper	To show at TRL 4
R3	Alarm is loud enough to wake the block	At least 85 dB(A) at 3 m	Sound level meter test	85.5 dB(A)	Met (est.)
R4	Alarm reaches neighbours and the block station	Every alarm in the block and the station within 10 s	Field test across a mock block layout	7.2 s worst case	Met (est.)
R5	Alarm runs without mains power	At least 12 months on batteries	Current draw measurement	Limited by cell shelf life, about 5 years	Met (est.)
R6	Cart can be moved on camp paths	Two adults move the loaded cart 100 m on an unpaved 10 % slope in 3 minutes or less; cart width 0.9 m or less	Timed trial on a test track	1.7 min; 101 N per person (1 N over the 100 N assumed sustainable); 850 mm wide	Met (est.)
R7	Hand pump delivers useful flow	At least 20 L/min (5.3 US gal/min) at the nozzle, pumped by two people for 10 minutes	Bench flow test with timed volume	20.4 L/min at 47 W per person	Met (est.)
R8	Jet reaches a burning shelter from a safe distance	Jet reach of 6 m (20 ft) or more	Measured throw test	6.9 m	Met (est.)
R9	Fast deployment	Water on target within 3 minutes of alarm for a shelter 100 m from the station	Timed drill with volunteers, from the siren, using the go-when-two-arrive rule	2.95 min (177 s) at 100 m; 2.5 min at 70 m	Met (est.), 3 s to spare at 100 m
R10	Repairable locally	All wear parts replaceable with hand tools and parts sold in local markets	Maintenance walkthrough with a camp workshop	Every wear part is a market item	Met by design
R11	Cost against the value-engineering target	Value-engineering target: USD 1,600 for one block kit	Costed bill of materials	USD 1,374, USD 226 under the target	Met

R9 stays at 3 minutes and 100 m. On 2026-10-03 Amish chose option A on R9 ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"; CBK-DDR-003): a check foot valve so the pump stays primed, the suction hose left coupled to the pump with a cam-lever coupling, and the drilled rule "go when two arrive" with a 30 s gather. That brought the 100 m time from 3.8 min to 3.2 min. The rest of the gap was the 26 s needed to fill the empty 30 m delivery hose. In round 2 Amish agreed to stow that hose full of water (CBK-DDR-004): "i agree with all the 46 recommendations you provided.

please proceed." The 100 m time is now 2.95 min. This is met on paper by only 3 s, on estimated inputs, and the delivery hose adds 8.9 kg to the cart: the pull is about 101 N per person on a 10 % slope, just over the 100 N assumed sustainable. The R6 time of 1.7 min is still met. The siting rule of CBK-DDR-001 (D9), no shelter more than 70 m along the path from a station, is kept as the siting margin; there the time is 2.5 min.

■ Assumptions

- A block has a drum or tap within reach of most shelters, or drums are placed at stations.
- Camp management allows alarms in shelters and a station in each block.
- Volunteers are trained and drill regularly, so the first two reach the station within 30 s of the siren.
- The foot valve's check holds the pump and suction hose full between uses, and the nozzle shut-off holds the delivery hose full; the weekly check on the drill card confirms both.
- The estimates marked (est.) in CBK-CAL-001 hold within the margins shown; the pump displacement is the most important one.

■ Safety

Safety: These requirements describe a pressurised hand-pumped water system and battery-powered alarms. The kit is for small, early fires only, people leave first, and water is never used on burning oil or live wiring. It is an open engineering reference, not certified equipment.